

Electronics

Solid-state diodes and diode circuits

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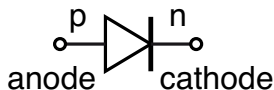
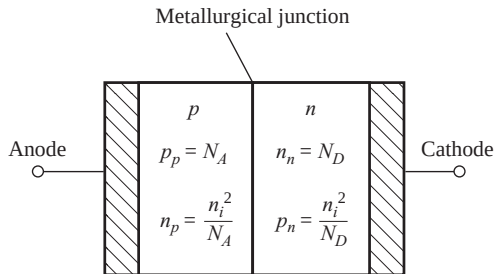
Outline

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- 4 Diodes under reverse bias
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The pn junction diode

The pn junction diode

- The pn junction diode is formed by fabrication of a p-type region in intimate contact with an n-type region
- The point at which the material changes from p-type to n-type is called the **metallurgical junction**
- The p-type region is the **anode** (doped with N_A atoms/cm³)
- The n-type region is the **cathode** (doped with N_D atoms/cm³)



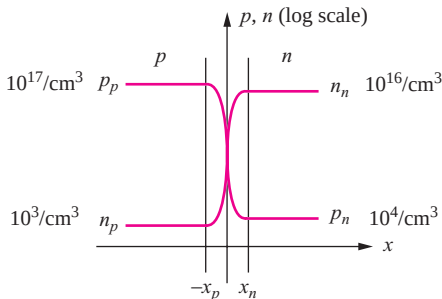
pn junction electrostatics

- Consider the diode doped with $N_A = 10^{17} \text{ cm}^{-3}$ on the p-type side and $N_D = 10^{16} \text{ cm}^{-3}$ on the n-type side
- Hole and electron concentrations are

$$\text{p-side: } p_p = N_A = 10^{17} \text{ cm}^{-3} \quad n_p = n_i^2 / N_A = 10^3 \text{ cm}^{-3}$$

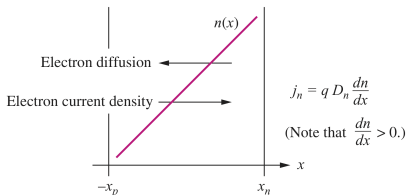
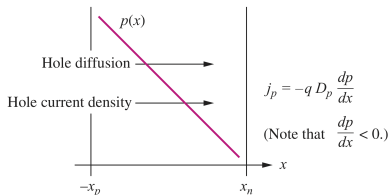
$$\text{n-side: } p_n = n_i^2 / N_D = 10^4 \text{ cm}^{-3} \quad n_n = N_D = 10^{16} \text{ cm}^{-3}$$

- A concentration gradient occurs at both sides of the metallurgical junction



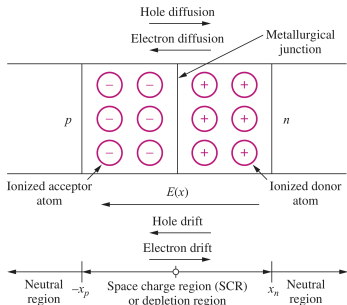
pn junction electrostatics

- Concentration gradients cause a diffusion current (due to both holes and electrons) from the p-type to the n-type region
- Electrons move from the n-type region to the p-type one
- They leave behind ionized donor atoms (fixed positive charges)
- In the p-type region they recombine with holes and create ionized acceptor atoms (fixed negative charges)
- The same holds for holes that moves from the p-type to the n-type region



pn junction electrostatics

- The process creates a number of fixed positive charges in the n-type region and the same number of fixed negative charges in the p-type one
- A **space charge region** or **depletion region** is created
- This space charge region induces an electric field and a **drift current** from the n-type region to the p-type one
- The equilibrium is reached when the diffusion current equals the drift current so that **the total current is zero**



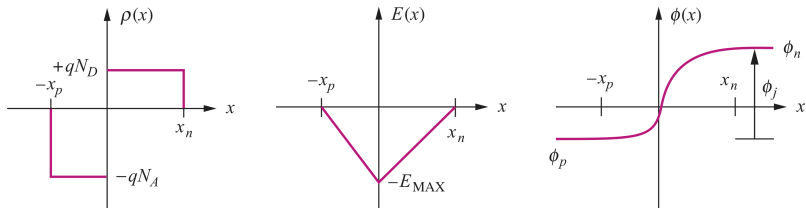
pn junction electrostatics

- A space charge region exists in the interval $[-x_p, x_n]$
- The charge neutrality imposes that

$$qN_A x_p = qN_D x_n$$

- From the charge density, $\rho_c(x)$ we determine the electric field

$$\nabla \cdot \mathcal{E} = \frac{\rho_c(x)}{\epsilon_s} \Rightarrow \mathcal{E}(x) = \frac{1}{\epsilon_s} \int \rho_c(x) dx$$



pn junction electrostatics

- From the electric field, we determine the electric potential as

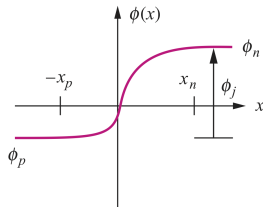
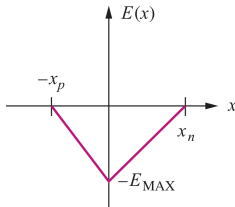
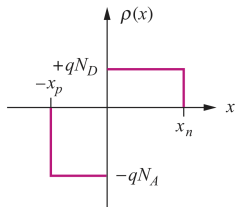
$$\phi(x) = - \int \mathcal{E}(x) dx$$

- A **junction potential** exists in $[-x_p, x_n]$

$$\phi_j = V_T \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

- The width of the depletion region results

$$w_{do} = x_p + x_n = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) \phi_j}$$



Internal diode current in equilibrium

- Let us consider the electrons but a similar discussion can be made for the holes
- Due to a concentration gradient a diffusion current exists

$$j_n^{\text{diff}} = qD_n \frac{\partial n}{\partial x}$$

- Since no current is measured at the output, the total current is zero

$$j_n^{\text{T}} = q\mu_n n \mathcal{E} + qD_n \frac{\partial n}{\partial x} = 0$$

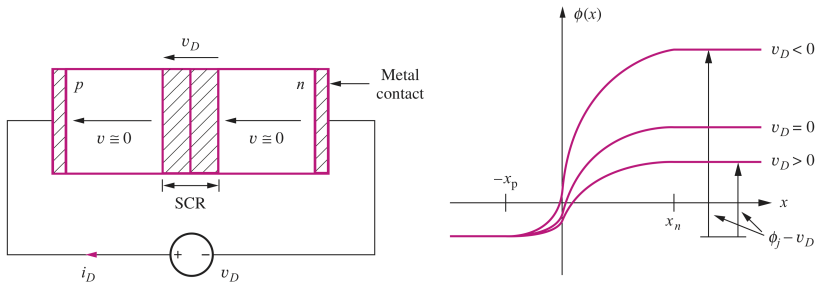
so that electron diffusion is exactly balanced by electron drift

- A drift current requires an electric field which is generated by the space charge region and produces a potential barrier, ϕ_j
- When a voltage is applied to the diode, the potential barrier is modified, and the delicate balances that makes $j^{\text{T}} = 0$ is disturbed, resulting in a current in the diode terminals

The i - v characteristics of the diode

The i - v characteristics of the diode

- Voltage, v_D , is applied from the anode to the cathode
- All the voltage, v_D , drops across the space charge region
- The potential barrier is modulated so that ϕ_j becomes $\phi_j - v_D$
- If $v_D > 0$ the potential barrier is reduced and a large diffusion current crosses the junction
- If $v_D < 0$ the potential barrier is increased and a negligible drift current prevails

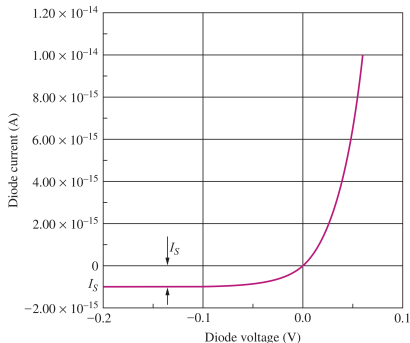


The i - v characteristics of the diode

- The current in a diode is a non-linear function of the voltage applied

$$i_D = I_S \left[\exp \left(\frac{v_D}{nV_T} \right) - 1 \right]$$

- I_S is the **reverse saturation current** (A)
- $V_T = kT/q$ is the thermal voltage (V)
- n is the **nonideality factor**



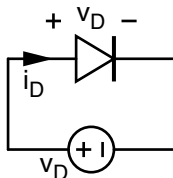
The i - v characteristics of the diode

- The diode saturation current is typically in the range

$$10^{-18} \text{ A} \leq I_S \leq 10^{-9} \text{ A}$$

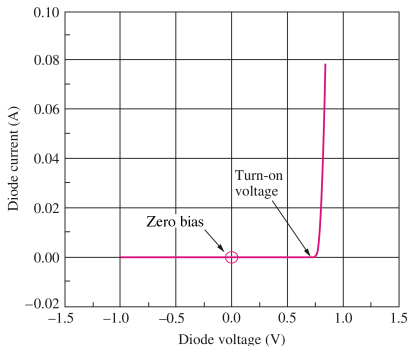
- I_S is proportional to n_i^2 and is strongly dependent on temperature
- For most silicon diodes, n is in the range 1.0 to 1.1 ($n \sim 2$ in diodes operating at high current densities)
- In our applications we assume $n = 1$ and the diode equation will be written as

$$i_D = I_S \left[\exp \left(\frac{v_D}{V_T} \right) - 1 \right]$$



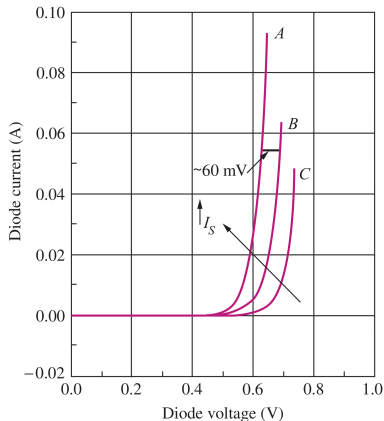
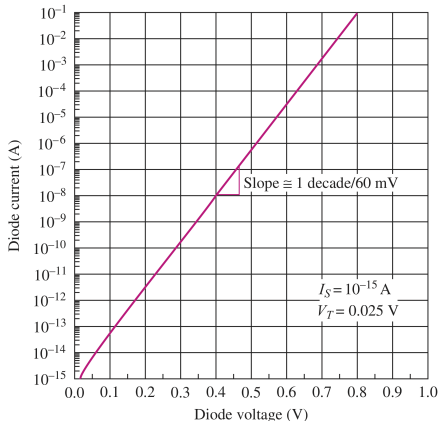
The i - v characteristics of the diode

- The diode characteristic is definitely not linear
- For $v_D < 0$ the diode is essentially nonconducting, with $i_D \sim 0$
- For $v_D > 0$ the current remains nearly zero until the voltage $v_D \sim 0.5$ – 0.7 V (turn-on voltage)
- At this point, the diode current increases rapidly, and the voltage across the diode becomes almost independent of current



The i - v characteristics of the diode

- A plot of the diode current in a semilogarithmic plot shows a clear straight line behavior
- Increasing I_S makes the turn-on voltage decrease ($I_{SA} = 10I_{SB} = 100I_{SC}$)



Diode equation approximations

- If $v_D < 0$ the diode is in **reverse bias** and the current equation can be approximated by

$$i_D = I_S \left[\exp \left(\frac{v_D}{V_T} \right) - 1 \right] \approx -I_S \quad \text{for } v_D < -4V_T$$

- If $v_D > 0$ the diode is in **forward bias** and the current equation can be approximated by

$$i_D = I_S \left[\exp \left(\frac{v_D}{V_T} \right) - 1 \right] \approx I_S \exp \left(\frac{v_D}{V_T} \right) \quad \text{for } v_D > 4V_T$$

- Obviously, if $v_D = 0$ then $i_D = 0$

Diode Temperature coefficient

Diode Temperature coefficient

- Solving $i_D = I_S [\exp(v_D/V_T) - 1]$ with respect to v_D we have

$$v_D = V_T \ln \left(\frac{i_D}{I_S} + 1 \right) = \frac{kT}{q} \ln \left(\frac{i_D}{I_S} + 1 \right) \approx \frac{kT}{q} \ln \left(\frac{i_D}{I_S} \right)$$

- Considering that $I_S \propto n_i^2$, it can be demonstrated that the derivative of v_D with respect to temperature yields

$$\frac{dv_D}{dT} = \frac{v_D - V_{GO} - 3V_T}{T}$$

being $V_{GO} = E_{GO}/q = 1.12 \text{ V}$ the bandgap voltage at $T = 0 \text{ K}$

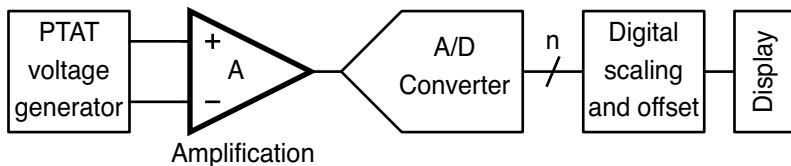
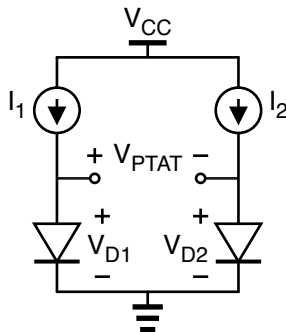
- Assuming a typical $v_D = 0.65 \text{ V}$ and $V_T = 0.025 \text{ V}$ at $T = 300 \text{ K}$, we have

$$\frac{dv_D}{dT} = \frac{0.65 - 1.12 - 3 \times 0.025}{300} = -1.82 \text{ mV/K}$$

Example: The PTAT voltage and electronic thermometry

- The Proportional-To-Absolute-Temperature (PTAT) voltage circuit forms the heart of most of today's highly accurate electronic thermometers
- Maintaining $I_1 = mI_2$ the output is independent of I_S

$$V_{PTAT} = V_{D1} - V_{D2} = \frac{kT}{q} \ln(m)$$



Diodes under reverse bias

Diodes under reverse bias

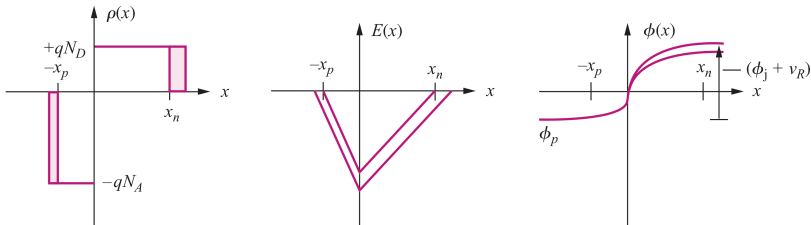
Space charge region width

- A reverse voltage v_R (positive at the cathode) applied to the diode adds directly to the built-in potential of the junction

$$\phi_j \rightarrow \phi_j + v_R$$

- The reverse voltage increases the width of the space charge region $w_d = x_p + x_n$

$$w_d = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (\phi_j + v_R)} = w_{do} \sqrt{1 + \frac{v_R}{\phi_j}}$$



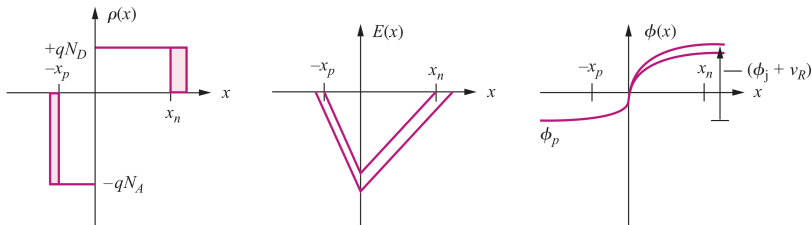
Diodes under reverse bias

Saturation current

- The reverse saturation current actually results from the thermal generation of hole-electron pairs in the depletion region
- It is therefore proportional to the volume of the depletion region, or

$$I_S = I_{S0} \sqrt{1 + \frac{V_R}{\phi_j}}$$

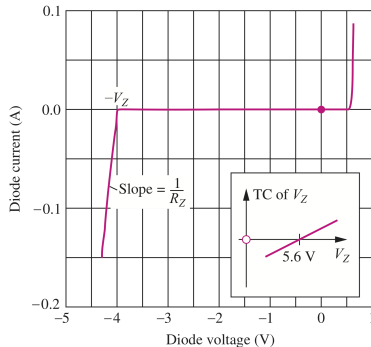
- Under forward bias, w_d changes very little, and $I_S = I_{S0}$ for forward bias



Diodes under reverse bias

Reverse breakdown

- As v_R increases, the electric field in the depletion region grows, and the diode eventually enters the **breakdown region**
- The current increases rapidly for any further increase in the applied voltage
- The voltage at which breakdown occurs is called the **breakdown voltage** V_Z ($2\text{ V} \leq V_Z \leq 2000\text{ V}$)
- Two breakdown mechanisms exist: **avalanche breakdown** and **Zener breakdown**



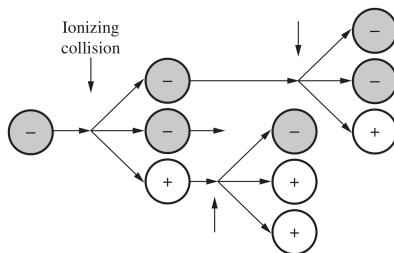
$$V_Z \propto \frac{1}{N}$$

N = doping level on the lightly doped side of the junction

Diodes under reverse bias

Avalanche breakdown

- In avalanche breakdown $V_Z > 5.6\text{ V}$
- As the width of the depletion layer increases under reverse bias, the electric field increases as well
- At some point, the electric field becomes large enough that some carriers gain energy sufficient to break covalent bonds upon impact, thereby creating electron-hole pairs
- The new carriers created can also accelerate and create additional electron-hole pairs through this impact-ionization process



Diodes under reverse bias

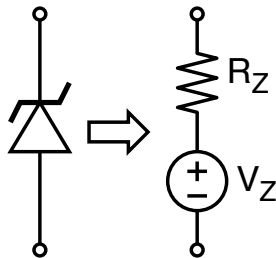
Zener breakdown

- True Zener breakdown occurs only in heavily doped diodes
- $V_Z < 5.6\text{ V}$
- The high doping results in a very narrow depletion-region width, and application of a reverse bias causes carriers to **tunnel** directly between the conduction and valence bands
- Avalanche and Zener breakdown have different temperature behavior
 - In avalanche breakdown V_Z increases with temperature
 - In Zener breakdown V_Z decreases with temperature

Diodes under reverse bias

Diode model for the breakdown region

- In breakdown, the diode can be modeled by a voltage source of value V_Z in series with resistor R_Z
- The value of R_Z is normally small ($R_Z \leq 100 \Omega$)
- The reverse current flowing in the diode must be limited by the external circuit or the diode will be destroyed
- Some diodes are actually designed to be operated in reverse breakdown and are called “Zener diodes” independently of the real breakdown mechanism



pn junction capacitance

pn junction capacitance

Reverse bias: junction capacitance

- Forward- and reverse-biased diodes have a capacitance associated with the pn junction
- These capacitive effects prevents the voltage across the diode from changing instantaneously
- In the reverse bias the capacitive effect is due to the charge accumulated in the depletion region, $Q_n = |Q_p|$
- Increasing v_R also the depletion charge increases

$$Q_n = qN_Dx_nA$$

- Substituting $N_Dx_n = N_Ax_p \Rightarrow x_p = (N_D/N_A)x_n$ into $w_d = x_p + x_n$ leads to

$$x_n = \frac{N_A}{N_D + N_A} w_d$$

- The depletion charge becomes

$$Q_n = q \frac{N_D N_A}{N_D + N_A} w_d A$$

pn junction capacitance

Reverse bias: junction capacitance

- Since $w_d = w_{do} \sqrt{1 + v_R / \phi_j}$, we have

$$Q_n = q \frac{N_D N_A}{N_D + N_A} w_{do} \sqrt{1 + \frac{v_R}{\phi_j}} A$$

- The junction capacitance is

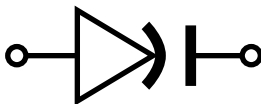
$$C_j = \frac{dQ_n}{dv_R} = \frac{C_{jo} A}{\sqrt{1 + \frac{v_R}{\phi_j}}}$$

- $C_{jo} = \frac{\epsilon_s}{w_{do}}$ represents the zero-bias junction capacitance per unit area of the diode

pn junction capacitance

Reverse bias: junction capacitance

- Diodes can be designed with impurity profiles (called hyper-abrupt profiles) specifically optimized for operation as voltage-controlled capacitors
- Remember that this diode is designed to be operated under reverse bias, but it conducts in the forward direction
- Circuit symbol for the variable capacitance diode (Varactor)



pn junction capacitance

Forward bias: diffusion capacitance

- When the diode is operating under forward bias, additional charge is stored in the neutral regions near the edges of the space charge region.
- The amount of charge Q_D stored in the diode is proportional to the diode current: $Q_D = i_D \tau_T$
- The proportionality constant τ_T is called the **diode transit time** and ranges from 1 fs to 1 μ s
- The diffusion capacitance C_D associated with the forward region of operation is

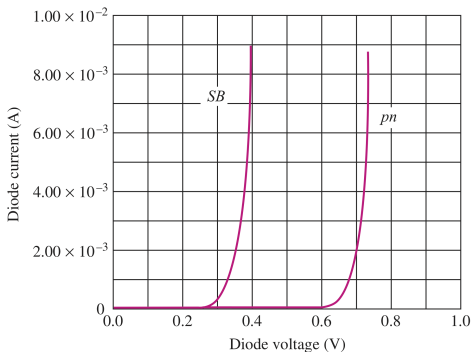
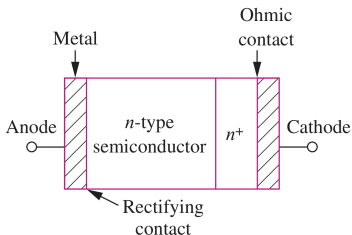
$$C_D = \frac{dQ_D}{dV_D} = \frac{(i_D + I_S)\tau_T}{V_T} \approx \frac{i_D \tau_T}{V_T}$$

- The diffusion capacitance is proportional to current and therefore can become quite large at high currents

Schottky barrier diode

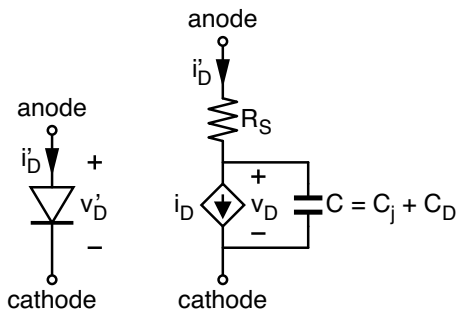
Schottky barrier diode

- In the Schottky diode, one of the semiconductor regions is replaced by a non-ohmic rectifying metal contact
- It is easiest to form a Schottky contact to n-type silicon, and for this case the metal region becomes the diode anode
- An n^+ region is added to ensure that the cathode contact is ohmic



Diode SPICE model

Diode SPICE model



$$i_D = IS \left[\exp \left(\frac{v_D}{N V_T} \right) - 1 \right]$$

$$C_D = TT \frac{i_D}{N V_T} \quad \text{for } v_D \geq 0$$

$$C_j = \frac{CJO}{\left(1 - \frac{v_D}{VJ} \right)^M} RAREA$$

Diode SPICE model

Parameter	Text	SPICE	Typ.
Saturation current	I_S	IS	10 fA
Series resistance	R_S	RS	0 Ω
Emission coefficient	n	N	1
Transit time	τ_T	TT	0 s
Zero-bias junction capacitance for a unit area RAREA = 1	$C_{j0} \cdot A$	CJO	0 F
Built-in potential	ϕ_j	VJ	1 V
Junction grading coefficient	—	M	0.5
Relative junction area	—	RAREA	1
Breakdown voltage	V_Z	BV	∞
Current at $V = BV$	—	IBV	10 nA
Breakdown emission coefficient	—	NBV	1

Diode SPICE model

■ D1N4002

Average Rectified Forward Current: 1 A

Non-Repetitive Peak Surge Current: 30 A

Peak Repetitive Reverse Voltage: 100 V

```
.MODEL D1N4002 D(IS=14.11N N=1.984 RS=33.89M IKF=94.81 XTI=3 EG=1.11  
+ CJO=51.17P M=.2762 VJ=.3905 FC=.5 ISR=100P NR=2 BV=100.1 IBV=10  
+ TT=4.761U)
```

■ D1N4148

Max continuous forward current: 200 mA

Max repetitive continuous reverse voltage: 100 V

```
.MODEL D1N4148 D(IS=2.682N N=1.836 RS=.5664 IKF=44.17M XTI=3 EG=1.11  
+ CJO=4P M=.3333 VJ=.5 FC=.5 ISR=1.565N NR=2 BV=100 IBV=100U TT=11.54N  
+ BV=5V)
```

■ ZENER D1N750

Zener Breakdown Voltage: 4.7 V

Maximum power dissipation: 500 mW

```
.MODEL D1N750 D(IS=880.5E-18 RS=.25 IKF=0 N=1 XTI=3 EG=1.11 CJO=175P  
+ M=.5516 VJ=.75 FC=.5 ISR=1.859N NR=2 BV=4.7 IBV=20.245M NBV=1.6989  
+ IBVL=1.9556M NBVL=14.976 TBV1=-21.277U)
```

Diode layout

